

# DRAMA: Debate-Driven Reasoning Agents for Matching Alignments

Yannick Oefler<sup>1,2,\*</sup>, Sophia Hammes<sup>1</sup>, Jan Portisch<sup>1</sup> and Sven Hertling<sup>2</sup>

<sup>1</sup>SAP SE, 69190 Walldorf, Germany

<sup>2</sup>University of Mannheim, 68131 Mannheim, Germany

## Abstract

Ontology matching involves correspondences of widely varying complexity: while many can be resolved through lexical or embedding-based similarity, ambiguous cases require deeper semantic reasoning—yet existing LLM-based approaches typically evaluate all candidate pairs in a single pass without explicit self-evaluation or iterative refinement. This paper presents DRAMA (**D**ebate-driven **R**easoning **A**gents for **M**atching **A**lignments), a multi-agent framework for ontology matching. The proposed approach is the first to combine structured multi-agent deliberation with critique-driven revision for ontology matching. The architecture employs an asymmetric actor–critic design in which a tool-augmented *Matching Agent* proposes correspondences, while a *Critic Agent* raises evidence-based objections that trigger iterative revision through a structured debate loop. A progressive pipeline cascades through stages of increasing reasoning complexity, ensuring that each mechanism is applied only to correspondences matching its reasoning requirements and allowing the system to scale to larger ontologies. Evaluated on the OAEI *Anatomy* and *Conference* tracks, DRAMA with GPT-5.5 achieves the highest  $F_1$ -scores on both benchmarks (0.951 and 0.688, respectively). These results are primarily driven by the highest recall among all evaluated systems. However, a precision gap remains compared with the best-performing structure-based matchers. These results provide initial evidence that debate-driven multi-agent architectures represent a viable research direction for ontology matching; at the same time, they reveal that underlying model capability currently contributes more to performance than the critic mechanism itself—raising the open question of whether an explicit critique step adds value when a sufficiently powerful model is employed

## Keywords

Ontology Matching, Large Language Models, Multi-Agent Debate, Actor–Critic, Tool-Augmented Reasoning

## 1. Introduction

Ontology matching—the task of identifying correspondences between semantically related entities across different ontologies [1]—is a prerequisite for achieving semantic interoperability across heterogeneous knowledge systems. However, the complexity of individual matching decisions varies substantially: while high-confidence correspondences can often be resolved through surface-level lexical or structural similarity, ambiguous cases require deeper semantic reasoning that integrates hierarchical context and the structural roles of entities within their respective ontologies.

Recent advances in large language models (LLMs) have enabled a new generation of ontology matching systems that employ an LLM as a downstream verifier, assessing candidate correspondences generated by conventional retrieval methods using zero-shot or few-shot prompting enriched with ontological context [2, 3, 4, 5, 6]. These systems achieve competitive performance on established benchmarks. However, they share a fundamental architectural limitation: candidate correspondences are evaluated in a single pass, without an explicit mechanism for critical self-evaluation or iterative revision when additional evidence becomes available.

Evidence from adjacent research domains shows that multi-agent debate [7, 8] and actor–critic architectures [9, 10] improve decision quality on complex reasoning tasks by introducing structured generate–evaluate cycles. Notably, most performance gains are typically achieved within only a few interaction rounds [11, 12, 13]. However, these advances have not yet been systematically transferred to ontology matching. Existing multi-agent approaches either lack iterative feedback mechanisms [14]

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✉ 0009-0001-6566-4698 (Y. Oefler); 0000-0001-5420-0663 (J. Portisch); 0000-0003-0333-5888 (S. Hertling)



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or constrain each agent to reasoning over a single ontology [15], leaving critique-driven iterative refinement largely unexplored.

This gap is particularly relevant in enterprise settings, where knowledge graph applications require reliable semantic correspondences across heterogeneous data sources at scale, yet established ontology matching systems often struggle with proprietary vocabularies and domain-specific terminology that lack coverage in public knowledge bases. Multi-agent architectures offer a promising direction for such knowledge-intensive tasks, as structured deliberation can surface implicit distinctions that single-pass approaches overlook.

This paper presents DRAMA (**D**ebate-driven **R**easoning **A**gents for **M**atching **A**lignments)—named for the deliberative conflict between its agents—an exploration of multi-agent deliberation for ontology matching. The system adopts a progressive architecture with stages of increasing reasoning complexity: embedding-based candidate generation, deterministic and lightweight LLM-based candidate resolution, and debate-driven matching. The final stage employs an asymmetric actor–critic architecture in which a tool-augmented *Matching Agent* explores ontological context through SPARQL-based retrieval and proposes correspondence decisions, while a *Critic Agent* evaluates the proposed correspondences, raises evidence-based objections, and triggers iterative revision. The main contributions of this work are:

1. **Debate-driven multi-agent matching:** We introduce DRAMA, a multi-agent ontology matching system that incorporates structured debate and critique-driven revision into the alignment process through an asymmetric actor–critic architecture with tool-augmented ontological reasoning.
2. **Progressive candidate resolution:** We design a staged architecture that combines embedding-based retrieval and lightweight classification to reduce the candidate space by 42–91% before multi-agent deliberation, limiting expensive tool-augmented reasoning to ambiguous cases.
3. **Comprehensive evaluation:** We evaluate DRAMA on the OAEI *Anatomy* and *Conference* tracks, providing a stage-wise analysis of pipeline components.

The remainder of this paper is organized as follows. Section 2 reviews related work on LLM-based ontology matching and multi-agent deliberation. Section 3 presents the DRAMA architecture and its four-stage pipeline, from ontology ingestion to debate-driven matching. Section 4 evaluates the proposed approach on two OAEI benchmarks and provides a stage-wise analysis of each pipeline component. Finally, Section 5 discusses the limitations of the current approach and outlines directions for future work.

## 2. Related Work

This work draws on three lines of research: LLM-based ontology matching, multi-agent architectures for ontology alignment, and critique-driven deliberation in NLP. The following sections review each of these areas and position DRAMA at their intersection. Table 1 compares existing systems across five architectural dimensions: number of agents, iterative revision, explicit critique, interaction paradigm, and ontological knowledge access. The comparison highlights the architectural gap addressed by DRAMA.

### 2.1. LLM-Based Ontology Alignment

The integration of large language models (LLMs) into ontology matching has advanced rapidly in recent years. Since prompting LLMs with entire ontologies does not scale beyond small matching tasks [2], most approaches employ the LLM as a downstream verifier that assesses candidate correspondences generated by conventional retrieval methods. Zero-shot and few-shot prompting achieves competitive performance on OAEI benchmarks when the LLM is provided with sufficient ontological context [3, 4, 5, 16]. To reduce inference cost, several approaches invoke the LLM only for low-confidence candidates, substantially improving efficiency without sacrificing alignment quality [6, 17].

**Table 1**

Architectural comparison of LLM-based ontology matching systems. Systems are grouped by architecture: single-agent (top), multi-agent without critique (middle), and multi-agent with critique (bottom). Columns summarise the number of LLM-based agents, support for iterative revision, the presence of an explicit critique mechanism, the interaction paradigm, and the ontological context available to the LLM.

| System             | Agents    | Iteration   | Critique | Paradigm          | Knowledge               |
|--------------------|-----------|-------------|----------|-------------------|-------------------------|
| Conv. OM [2]       | 1         | single-pass | ✗        | direct            | full ontology           |
| LLMs4OM [3]        | 1         | single-pass | ✗        | pipeline          | labels + hierarchy      |
| OLaLa [4]          | 1         | single-pass | ✗        | pipeline          | labels                  |
| LLMap [5]          | 1         | single-pass | ✗        | pipeline          | labels + hierarchy      |
| KROMA [16]         | 1         | single-pass | ✗        | pipeline          | SPARQL + ext. KBs       |
| MILA [17]          | 1         | single-pass | ✗        | pipeline          | labels                  |
| MapperGPT [18]     | 1         | single-pass | ✗        | pipeline          | definitions + hierarchy |
| SMATCH-M-LLM [19]  | 1         | single-pass | ✗        | pipeline          | ontology fragments      |
| LogMapLLM [21]     | 1         | single-pass | ✗        | pipeline          | labels + hierarchy      |
| Agent-OM [14]      | 2 (asym.) | single-pass | ✗        | pipeline decomp.  | full context            |
| LLMA Dialogue [15] | 2 (sym.)  | multi-pass  | ✗        | negotiation       | embeddings              |
| <b>DRAMA</b>       | 2 (asym.) | multi-pass  | ✓        | generate-evaluate | SPARQL + struct.        |

The contextual information supplied to the LLM varies across systems and includes hierarchical neighbourhoods [3, 4, 5, 6], lexical definitions [18], iteratively sampled ontology fragments [19], SPARQL-derived structural patterns [20], and combinations thereof enriched with external knowledge bases [16].

Despite these advances, existing approaches share a fundamental architectural limitation: they evaluate candidate correspondences in a single pass, without an explicit mechanism for critical assessment or revisiting previous decisions (see Table 1, top group). DRAMA addresses this limitation by introducing iterative critique, where proposed correspondences are challenged through evidence-based objections and refined through subsequent revisions rather than finalized after a single LLM invocation.

Furthermore, whereas prior systems typically provide the LLM with a fixed context assembled before inference, DRAMA’s Matching Agent actively retrieves additional ontological evidence during deliberation through SPARQL-based queries.

## 2.2. Multi-Agent Approaches for Ontology Alignment

Recent work has started to investigate multi-agent architectures for ontology alignment, but existing approaches provide only limited forms of agent interaction. AGENT-OM [14] decomposes the matching task across two sequential LLM agents without iterative feedback, resulting in a pipeline-based architecture rather than a deliberative process. Zhang et al. [15] introduce iterative interaction through a symmetric negotiation protocol. By construction, neither agent has simultaneous access to both ontologies and instead relies on embedding similarity and peer proposals to identify matches. Performance on the OAEI Anatomy track (66.6%  $F_1$ -score) remains below state-of-the-art, suggesting that restricted information access limits alignment quality.

Table 1 contrasts existing LLM-based ontology matching systems across five dimensions. The comparison highlights a clear gap: no existing approach combines multi-agent deliberation with autonomous tool use, iterative revision, and an explicit critique mechanism.

DRAMA addresses this gap through a *debate-driven* multi-agent architecture comprising two asymmetric agents: a *Matcher* that proposes entity-level correspondences and a *Critic* that evaluates the evolving alignment and raises evidence-based objections. Disputed correspondences are returned to the *Matcher* for revision, forming an iterative deliberation loop that incrementally refines the alignment.

### 2.3. Multi-Agent Deliberation and Critique in NLP

The debate-driven *generate-evaluate* paradigm adopted in this work is well supported by recent advances in NLP. Multi-agent debate has been shown to improve factuality and reasoning quality across a range of tasks [7, 22, 23, 24], while theoretical analyses highlight agent diversity as a necessary condition for avoiding convergence toward shared misconceptions [8]. These effects are particularly pronounced in zero-shot settings [11]—the regime in which ontology matching typically operates—and are primarily driven by iterative belief revision in response to opposing viewpoints rather than repeated independent generation [25].

Within this broader paradigm, asymmetric architectures that decouple generation and evaluation have proven especially effective. Jointly trained *actor-critic* systems outperform both single-agent and symmetric debate baselines [9], while lightweight critics can effectively supervise substantially larger generators, reflecting a fundamental generation-verification asymmetry that generalises across domains including design and dialogue generation [10, 26, 27]. In addition, evidence indicates that critique quality is a key determinant of performance: deliberate, multi-perspective evaluation consistently outperforms shallow LLM feedback for both error detection and iterative refinement [12, 13, 28], and most gains from deliberation are achieved within a small number of interaction rounds [29].

Collectively, these findings motivate the design of DRAMA: an asymmetric multi-agent architecture with complementary roles, bounded iterative deliberation via a structured feedback loop, and a debate-driven interaction mode in which the *Critic* produces evidence-based objections.

## 3. DRAMA

DRAMA aligns a source ontology  $O_s$  with a target ontology  $O_t$  by identifying equivalence correspondences between ontology classes; property alignment is outside the scope of this work. As illustrated in Figure 1, the architecture consists of four components that form a progressive reasoning cascade [6, 17, 30, 31], applying increasingly sophisticated mechanisms only to candidates requiring deeper analysis.

*Ontology Ingestion* (OI) parses both ontologies and stores structured entity representations in a shared database. *Candidate Generation* (CG) performs embedding-based  $k$ -nearest-neighbour retrieval to produce ranked candidate pairs [1, 4, 30, 32]. *Candidate Resolution* (CR) resolves straightforward cases through score-based filtering and lightweight LLM classification. The remaining ambiguous candidates are processed by *Debate-Driven Matching* (DDM), where a *Matcher* agent proposes correspondences and a *Critic* agent iteratively refines them through a structured propose-critique-revise loop.

The remainder of this section describes each component in detail. Sections 3.1–3.3 present the preprocessing and candidate resolution pipeline, while Section 3.4 introduces the debate-driven matching framework, which constitutes the primary contribution of this work.

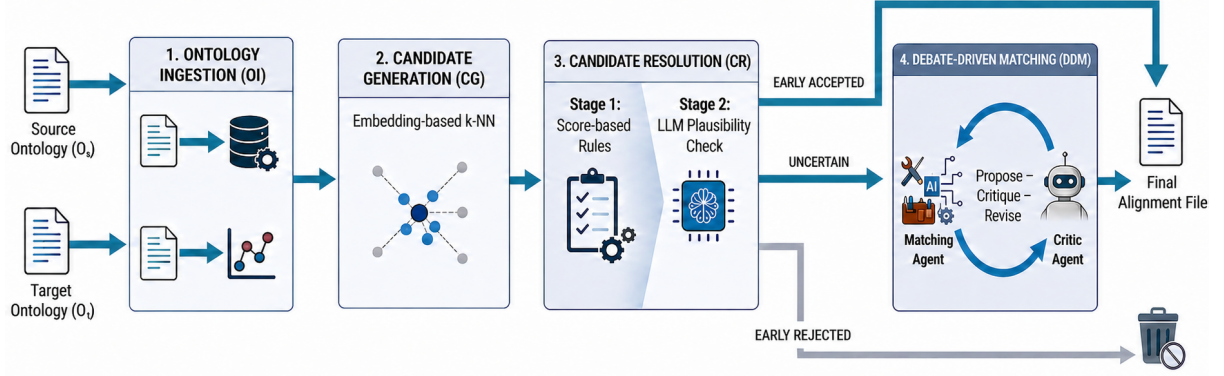
### 3.1. Ontology Ingestion (OI)

OI parses the input ontologies  $O_s$  and  $O_t$  and constructs an entity-centric representation for each class: URI fragment, labels, textual definitions, and direct superclasses with their preferred labels [3, 4, 5]. URI fragments containing more than 50% numeric characters are discarded [4]. Entity descriptions are further enriched through recursive traversal of annotation-property links.

The resulting representations are stored in a relational database for access by CG and CR, while the complete ontology graphs are loaded into an embedded SPARQL triplestore to support multi-hop traversal during DDM.

### 3.2. Candidate Generation (CG)

CG reduces the quadratic search space to a manageable candidate set while maintaining high recall. Each textual element extracted by OI—labels, definitions, and URI fragments—is embedded independently



**Figure 1:** Overview of the DRAMA pipeline. The four stages apply progressively more expensive reasoning: embedding-based retrieval (CG), deterministic rules and lightweight LLM classification (CR), and tool-augmented multi-agent deliberation (DDM). Accepted correspondences from CR and DDM are aggregated into the final alignment.

using a dense embedding model, resulting in a variable-length set of embeddings per entity rather than a single aggregated representation [4]. This preserves the semantic signal of individual labels and allows synonyms to match directly without dilution.

Candidate retrieval is performed bidirectionally: for each entity in  $O_s$ , the  $k$  most similar entities in  $O_t$  are retrieved, and vice versa. Text-level similarities are aggregated into entity-level scores via max-pooling over all text pairs. If a pair is retrieved from both directions, both retrieval outcomes are retained independently. The embedding model and  $k$  are treated as configurable hyperparameters evaluated in Section 4.

### 3.3. Candidate Resolution (CR)

Candidate Resolution (CR) bridges CG and DDM through a two-stage cascade. It resolves unambiguous correspondences using progressively more expressive decision mechanisms while forwarding only genuinely ambiguous cases to the computationally expensive multi-agent stage. This design follows the principle that reasoning effort should increase with decision difficulty [6, 17].

**Stage 1: Score-based resolution.** Stage 1 applies deterministic decision rules based exclusively on the similarity scores and retrieval directions produced by CG. Candidate pairs whose similarity falls below the lower threshold  $\tau_{\text{low}}$  are discarded immediately:

$$\text{sim\_score} < \tau_{\text{low}} \implies \text{REJECT}.$$

The remaining candidates are evaluated for direct acceptance using conservative criteria that depend on the retrieval configuration.

**For  $k = 1$ .** A candidate is accepted only if both entities are each other’s highest-ranked retrieval result (i.e., mutual nearest neighbours) and their similarity score exceeds the upper threshold:

$$\text{sim\_score} \geq \tau_{\text{high}}.$$

**For  $k > 1$ .** Since multiple candidates compete for the same source entity, a pair  $(e_s, e_t)$  is accepted only if it satisfies three conditions: (i) mutual nearest-neighbour status, (ii)  $\text{sim\_score} \geq \tau_{\text{high}}$ , and (iii) a minimum separation margin of at least  $\delta$  between the top-ranked and second-ranked candidate, computed independently for each entity in the pair. Formally, let  $c_1$  and  $c_2$  denote the highest- and second-highest-scoring candidates for a given entity; then:

$$\text{margin}(e_s) = \text{sim\_score}(e_s, c_1) - \text{sim\_score}(e_s, c_2) \geq \delta,$$

and analogously for  $e_t$ . If only one candidate remains after pruning, the margin condition is vacuously satisfied.

All accepted pairs must satisfy a conflict-aware one-to-one constraint: if multiple eligible pairs compete for the same entity, none is accepted and all are forwarded to Stage 2; conversely, candidates involving an already-assigned entity are rejected. While this simplifies over the small number of m:n correspondences in both reference alignments, the separation margin  $\delta$  ensures that ambiguous cases are deferred rather than committed prematurely.

**Stage 2: LLM plausibility check.** Stage 2 receives all unresolved candidate pairs from Stage 1, supplemented by structured entity context (labels, descriptions, superclass labels) from the OI database. For each source entity, all associated target candidates are presented jointly within a single prompt, ordered by decreasing similarity score to exploit the primacy bias of autoregressive models [33]. The model generates verdicts using constrained decoding [34] over the label set  $\mathcal{L} = \{\text{ACCEPT}, \text{REJECT}, \text{UNCERTAIN}\}$ . Since generation is autoregressive, each prediction is conditioned on previously generated verdicts. Label log-probabilities are length-normalised and converted to confidence estimates via softmax.

A dual-threshold gating mechanism commits a verdict only if its confidence exceeds the label-specific threshold ( $\alpha_{\text{accept}}$  or  $\alpha_{\text{reject}}$ ); predictions below threshold are mapped to UNCERTAIN. Classification is performed independently from both retrieval directions. The directional verdicts are merged conservatively: a REJECT in either direction eliminates the pair unless contradicted by an explicit ACCEPT from the opposite direction, in which case the pair is marked UNCERTAIN. A final ACCEPT requires bilateral agreement; all other combinations default to UNCERTAIN.

### 3.4. Debate-Driven Matching (DDM)

The DDM phase processes candidate correspondences that remain unresolved after CR and produces the final alignment through structured deliberation. The architecture follows an asymmetric actor–critic design: a tool-augmented *Matching Agent* autonomously explores ontological context and proposes alignment decisions, while a *Critic Agent* evaluates these decisions and raises objections to insufficiently justified matches. Objections are fed back to the *Matching Agent*, forming an iterative propose–critique–revise cycle that forms the core debate mechanism of the system.

The *Matching Agent* is implemented as a ReAct [35] agent graph using LangGraph [36], with tool access standardised through MCP [37]. Decisions and objections are persisted in the shared pipeline database, which serves as both the communication channel between agents and the authoritative source for alignment extraction. This implements database-backed symbolic memory [38]: the agent retrieves prior decisions on demand via tool calls, keeping its context bounded regardless of corpus size.

**Matching Agent.** The Matching Agent is invoked once per source entity and receives pre-loaded contextual information—preferred labels, descriptions, and superclass labels—for both the source entity and all candidate targets, providing an immediate point of orientation without requiring initial tool calls. The agent evaluates each candidate pair, optionally collecting additional ontological evidence, and assigns a verdict from  $\{\text{MATCH}, \text{NO\_MATCH}\}$  together with a natural-language justification. Five tools are available to the agent, grouped according to their functionality:

1. **Retrieval evidence:** Provides the embedding cosine similarities produced during the CG stage.

2. **Ontology traversal:** Enables arbitrary SPARQL queries against the triplestore, giving the agent unrestricted access to the full ontological structure beyond the pre-loaded context.
3. **Code execution:** Executes short Python snippets for auxiliary operations.
4. **Short-term memory:** Provides a key-value scratchpad for intermediate findings. The agent records insights after each ontology query and retrieves the accumulated evidence before finalizing decisions, decoupling evidence accumulation from context-window constraints.
5. **Decision persistence:** Persists all decisions for the current source entity in the shared database.

The availability of SPARQL-based traversal distinguishes this approach from prior systems that rely exclusively on prompt-injected context [2, 4, 5], enabling the agent to autonomously determine *which* information to retrieve as its understanding of a candidate pair evolves.

A limitation of the current design is that source entities are processed sequentially without backtracking: if a stronger correspondence for an already-assigned target is identified later, the earlier decision cannot be retracted. Extending the framework with such a revision mechanism remains an important direction for future work.

**Critic Agent.** The *Critic Agent* is invoked for every MATCH decision. It receives the source and target entity metadata, the *Matching Agent*’s verdict and justification, and a summary of the tools used during reasoning. Unlike the *Matching Agent*, the *Critic* has no access to external tools and evaluates only the consistency, plausibility, and evidential sufficiency of the proposed justification, thereby enforcing a clear separation between evidence acquisition and evidence evaluation. It returns one of two verdicts, {AGREE, OBJECT}, together with a natural-language justification.

If the *Critic* objects, its rationale is appended to the *Matching Agent*’s conversation and the agent is prompted to revise. This cycle repeats for up to  $n$  revision rounds; once the limit is reached or the *Critic* agrees, the latest verdict is retained as final.

## 4. Evaluation

DRAMA is evaluated on two established benchmarks from the Ontology Alignment Evaluation Initiative (OAEI) [39]: the *Anatomy* and *Conference* tracks. We first report end-to-end alignment quality in Section 4.1, then analyse each pipeline stage individually in Sections 4.2–4.4.

Embedding computation for the Candidate Generation stage and inference of the Candidate Resolution classifier were performed on a single NVIDIA RTX 6000 Ada GPU (48 GB VRAM). The DDM stage employed foundation models accessed through an API.

### 4.1. End-to-End Results

Table 2 presents the end-to-end performance of DRAMA, combining correspondences accepted during the Candidate Resolution (CR) stage with those produced by Debate-Driven Matching (DDM). Results are compared against state-of-the-art ontology matching systems evaluated in the OAEI 2025 campaign. As DRAMA’s development commenced in April 2026, after the OAEI 2025 submission deadline had passed, it did not participate in the official evaluation; results for all comparison systems are therefore taken directly from the official OAEI 2025 results pages [40, 41].

On the *Anatomy* track, DRAMA with GPT-5.5 achieves the best overall performance, obtaining an  $F_1$ -score of 0.951 and surpassing Matcha (0.941) and Agent-OM (0.920). It also achieves the highest recall (0.961) among all evaluated systems, while maintaining a precision of 0.944, exceeded only by LogMapLLM (0.964). Even when using the open-weight Mistral Small model, DRAMA attains an  $F_1$ -score of 0.912, matching ALIN and outperforming LogMapLLM.

On the *Conference* track, DRAMA with GPT-5.5 again achieves the highest  $F_1$ -score (0.688), outperforming all comparison systems, including ALIN (0.650), LogMap (0.640), Matcha (0.630), and Agent-OM (0.610). This improvement is primarily driven by recall, which reaches 0.734 and exceeds the next-best system (ALIN, 0.680). Precision (0.647), however, remains below that of structure-based matchers such

**Table 2**

End-to-end results on the OAEI *Anatomy* and *Conference* tracks. Comparison systems correspond to the four top-performing matchers and the string equivalence baseline from the OAEI 2025 campaign. Their results are taken from the official OAEI 2025 results pages [40, 41] and retain the reported decimal precision. For the *Conference* track (21 ontology pairs), Precision, Recall, and  $F_1$  are micro-averaged. Size denotes the number of predicted correspondences; — indicates unavailable data.

| System                        | Precision    | Recall       | F1           | Size  |
|-------------------------------|--------------|--------------|--------------|-------|
| Anatomy                       |              |              |              |       |
| DRAMA (Mistral Small, $k=5$ ) | 0.898        | 0.926        | 0.912        | 1 562 |
| DRAMA (GPT-5.5, $k=5$ )       | 0.944        | <b>0.961</b> | <b>0.951</b> | 1 541 |
| StringEquiv                   | <b>0.997</b> | 0.622        | 0.766        | 946   |
| LogMapLLM [6]                 | 0.964        | 0.842        | 0.899        | 1 324 |
| Agent-OM [14]                 | 0.959        | 0.883        | 0.920        | 1 396 |
| Matcha [42]                   | 0.951        | 0.931        | 0.941        | 1 485 |
| ALIN [43]                     | 0.942        | 0.884        | 0.912        | 1 423 |
| Conference                    |              |              |              |       |
| DRAMA (Mistral Small)         | 0.64         | 0.60         | 0.62         | 286   |
| DRAMA (GPT-5.5)               | 0.65         | <b>0.73</b>  | <b>0.69</b>  | 346   |
| StringEquiv                   | 0.76         | 0.41         | 0.53         | —     |
| Agent-OM [14]                 | 0.64         | 0.59         | 0.61         | —     |
| LogMap [30]                   | 0.76         | 0.56         | 0.64         | —     |
| Matcha [42]                   | <b>0.77</b>  | 0.53         | 0.63         | —     |
| ALIN [43]                     | 0.62         | 0.68         | 0.65         | —     |

as Matcha (0.770) and LogMap (0.760), suggesting that the tendency toward over-matching persists on structurally shallow ontologies, albeit to a lesser extent than in the DDM-only evaluation.

Across both benchmarks, DRAMA exhibits a recall-oriented performance profile, consistently achieving the highest recall while maintaining competitive, but not state-of-the-art, precision. One possible explanation is the exploratory nature of the proposed architecture: the tool-augmented *Matching Agent* actively gathers supporting evidence through iterative exploration of the ontologies, potentially placing less emphasis on evidence that contradicts a proposed correspondence.

## 4.2. Candidate Generation

The CG stage determines the upper bound on end-to-end recall, as correspondences not retrieved at this point cannot be recovered by subsequent stages. This section evaluates the retrieval recall and candidate set size produced by bidirectional  $k$ -nearest-neighbour retrieval (Table 3).

KaLM-Embedding-Gemma3-12B-2511 was selected based on a systematic ablation study comprising seven models from 110M to 12B parameters (Appendix A, Table 8). To validate our retrieval implementation, we reproduced the candidate generation recall reported by OLaLa [4] under identical settings using their embedding model. Scaling to KaLM-12B reduces irrecoverable false negatives on the *Anatomy* track from 34 to 11 at  $k=5$ .

On the *Anatomy* track, retrieval achieves near-complete coverage at  $k=5$  (11 of 1 516 correspondences missed), establishing a tight upper bound for downstream matching. On the *Conference* track, recall reaches 0.827 at  $k=5$ —lower than *Anatomy*, as the short and abstract labels of *Conference* ontologies provide fewer discriminative signals for embedding-based retrieval. With 55 of 305 reference correspondences unrecovered, the maximum achievable recall of subsequent stages is bounded at approximately 0.82; recovering these cases will require complementary retrieval strategies left for future work.

Based on these results, we adopt  $k=5$  for all subsequent *Conference* experiments, as the retrieval recall of 0.735 at  $k=1$  was considered insufficient to justify running the full downstream pipeline under this configuration. For the *Anatomy* track, recall at  $k=1$  (0.960) already approaches the ceiling achieved at

**Table 3**

CG performance using bidirectional  $k$ -nearest-neighbour retrieval with KaLM-Embedding-Gemma3-12B-2511 on the OAEI *Anatomy* and *Conference* tracks for retrieval depths  $k=1$  and  $k=5$ . *FN* reports the number of missed reference correspondences. *Size* indicates the total number of candidates forwarded to the CR stage. For the *Conference* track (21 ontology pairs), Recall is macro-averaged across pairs, while FN, Size, and Runtime are aggregated sums.

| Track      | $k = 1$ |    |       |             | $k = 5$ |    |        |             |
|------------|---------|----|-------|-------------|---------|----|--------|-------------|
|            | Recall  | FN | Size  | Runtime (s) | Recall  | FN | Size   | Runtime (s) |
| Anatomy    | 0.960   | 60 | 4 560 | 263.36      | 0.993   | 11 | 22 540 | 282.78      |
| Conference | 0.735   | 85 | 3 285 | 1 566.00    | 0.827   | 55 | 15 747 | 513.30      |

$k=5$  (0.993); we therefore retain this configuration as a secondary evaluation setting to analyse how the reduced candidate set affects downstream workload and computational cost, in particular within the DDM stage.

### 4.3. Candidate Resolution

CR is evaluated in a stage-wise manner. Stage 1 applies deterministic score-based filtering and matching, whereas Stage 2 employs a lightweight LLM classifier to resolve the remaining candidate pairs. The evaluation focuses on the reduction of the candidate set size, the resulting upper bound on achievable recall, and the precision of correspondences resolved during these early stages. All CR hyperparameters are tuned exclusively on the *Anatomy* track; the *Conference* track serves as a zero-shot transfer setting to assess the out-of-domain generalisability of the selected configuration.

**Stage 1: Score-based filtering and matching.** Stage 1 applies two deterministic rules in sequence: (i) pruning candidate pairs below a similarity threshold  $\tau_{\text{low}}$ , and (ii) accepting candidates exceeding a higher threshold  $\tau_{\text{high}}$  with sufficient separation from the next-best candidate. The thresholds  $\tau_{\text{low}}=0.71$ ,  $\tau_{\text{high}}=0.97$ , and  $\delta=0.01$  were selected based on ablation studies on the *Anatomy* track (Appendix B, Tables 9–11). The resulting performance is summarised in Table 4.

On the *Anatomy* track, pruning retains all reference correspondences by design, as  $\tau_{\text{low}}$  was calibrated to achieve zero gold correspondence loss. Applied without additional tuning to the *Conference* track, the same threshold removes 58.9% of candidate pairs while preserving 98.8% of reference correspondences, indicating effective cross-track generalisability of the retrieval filtering stage.

The acceptance rules achieve perfect precision on *Anatomy* at  $k=5$ , but only 85.3% micro-precision on *Conference*, showing that the configuration calibrated on *Anatomy* does not fully transfer to structurally heterogeneous benchmarks. A more conservative acceptance strategy, for example through a higher  $\tau_{\text{high}}$  or an increased separation margin, may provide a more favourable precision–coverage trade-off.

Furthermore, Stage 1 enforces a one-to-one matching constraint by discarding candidates whose target entity has already been assigned to a higher-scoring accepted correspondence (reported as “Lock-rej.” in Table 4). Combining score-based pruning and acceptance, Stage 1 reduces the candidate space by 72.75% for the *Conference* track and by 20.79% for the *Anatomy* track ( $k=5$ ) before any LLM inference is performed. The remaining candidates—4 291 pairs for *Conference* and 17 853 for *Anatomy* ( $k=5$ )—are forwarded to Stage 2 for subsequent LLM-based classification.

**Stage 2: LLM plausibility check.** Following Stage 1, the remaining candidate pairs are processed by a lightweight LLM classifier—Qwen2.5-7B-Instruct [44]—that assigns one of three labels (*accept*, *reject*, or *uncertain*) based on entity labels, descriptions, and embedding similarity scores. The classifier operates with confidence thresholds  $\alpha_{\text{accept}}=0.9$  and  $\alpha_{\text{reject}}=0.8$ . Both thresholds were determined by analysing the confidence-score distribution on the *Anatomy* track at  $k=1$  and selecting the most

**Table 4**

CR Stage 1 results combining pruning ( $\tau_{\text{low}}=0.71$ ) and acceptance ( $\tau_{\text{high}}=0.97$ ,  $\delta=0.01$  for  $k>1$ ) for *Anatomy* and *Conference* (at  $k=5$ ). *Accepted* denotes pairs committed as matches (TP+FP); *Prec.* is the precision of these accepted pairs. *Lock-rej.* denotes candidates excluded by the one-to-one constraint after a higher-scoring match binds their target entity. *Total Red.* is the cumulative size reduction relative to CG output. For *Conference* (21 pairs), Recall and Precision are micro-averaged; remaining columns are sums.

| Track             | Pruning ( $\tau_{\text{low}}$ ) |        |           | Acceptance ( $\tau_{\text{high}}$ ) |          |       | Stage 1 Output |            |             |
|-------------------|---------------------------------|--------|-----------|-------------------------------------|----------|-------|----------------|------------|-------------|
|                   | Pruned                          | Recall | Reduction | Accepted                            | Rejected | Prec. | Size           | Total Red. | Runtime (s) |
| Anatomy ( $k=1$ ) | 201                             | 1.000  | 4.41%     | 101                                 | 505      | 0.980 | 3 753          | 17.70%     | 3.00        |
| Anatomy ( $k=5$ ) | 2 476                           | 1.000  | 10.98%    | 98                                  | 2 113    | 1.000 | 17 853         | 20.79%     | 5.02        |
| Conference        | 9 281                           | 0.988  | 58.94%    | 163                                 | 2 012    | 0.853 | 4 291          | 72.75%     | 4.26        |

**Table 5**

CR Stage 2 results using Qwen2.5-7B-Instruct ( $\alpha_{\text{accept}}=0.9$ ,  $\alpha_{\text{reject}}=0.8$ ). *Reject* includes lock rejections from the one-to-one constraint. *Total Red.* is cumulative relative to CG output; *Size* is the number of pairs forwarded to DDM. Conference metrics are micro-averaged across 21 ontology pairs, and all remaining columns report aggregated sums.

| Track             | Accept | Accept Prec. | Reject | Reject Prec. | Total Red. | Size  | Runtime (s) |
|-------------------|--------|--------------|--------|--------------|------------|-------|-------------|
| Anatomy ( $k=1$ ) | 43     | 1.000        | 1,046  | 0.992        | 41.64%     | 2 664 | 1 172       |
| Anatomy ( $k=5$ ) | 1 118  | 0.979        | 14 727 | 0.998        | 91.11%     | 2 008 | 7 531       |
| Conference        | 16     | 0.625        | 2 894  | 0.997        | 91.17%     | 1 382 | 1 621       |

aggressive values that maintained error-free precision on this tuning configuration. Table 5 summarises the results. More capable LLMs may further improve the classifier’s precision–coverage trade-off, an aspect left for future work.

The classifier’s primary contribution is the reliable identification of non-matching candidate pairs. Reject precision ranges from 0.992 to 0.998 across all evaluated configurations, enabling aggressive candidate pruning with virtually no false rejections. In contrast, acceptance performance is considerably weaker, particularly on the *Conference* track, where micro-averaged acceptance precision reaches only 62.5%. This indicates that the acceptance threshold calibrated on *Anatomy* does not transfer reliably to structurally different ontologies. The resulting asymmetry confirms the intended division of labour: the lightweight classifier can reliably eliminate clear non-matches but lacks the semantic reasoning capability required to confirm equivalence in more ambiguous cases.

At  $k=5$ , the one-to-one constraint further amplifies the effect of the classifier through lock rejections, but also removes 18 reference correspondences on the *Anatomy* track. After Stage 2, fewer than 9% of the original candidate pairs remain for both the *Anatomy* and *Conference* tracks, representing the subset of candidate pairs forwarded to the DDM stage for further deliberation.

#### 4.4. Debate-Driven Matching

Table 6 reports the isolated performance of DDM, evaluated exclusively on candidate pairs classified as UNCERTAIN by CR. This constitutes a more challenging setting than end-to-end evaluation, as all straightforward correspondences have already been resolved by the preceding stages. Two language models were evaluated as the *Matching Agent*: Mistral Small 4 [45] (119B parameters), an open-weight model, and GPT-5.5 [46]. The *Critic Agent* employs Mistral Small 4 in all configurations; evaluating alternative critic models and configurations is left for future work.

On the *Anatomy* track, GPT-5.5 at  $k=1$  achieves  $F1 = 0.965$  ( $P = 0.951$ ,  $R = 0.978$ ), substantially outperforming Mistral Small under the same configuration ( $F1 = 0.859$ ,  $P = 0.800$ ,  $R = 0.928$ ). At  $k=5$ , the larger candidate set reduces precision for both models, with Mistral Small dropping to  $F1 = 0.669$  and GPT-5.5 to  $F1 = 0.866$ . Performance on the *Conference* track is substantially lower across all configurations, with

**Table 6**

DDM stage performance evaluated exclusively on candidate pairs classified as UNCERTAIN by CR. Results compare different LLMs and the effect of Critic-based revision. *Objections* denotes the number of correspondences challenged by the *Critic*; *Flips* denotes decisions that changed from MATCH to NO\_MATCH after revision (both inapplicable for the Base variant, which operates without *Critic*). Metrics for the *Conference* track (21 ontology pairs) are micro-averaged.

| Model         | $k$ | Variant  | Precision | Recall | F1    | Tokens (M) | Runtime (h) | Objections | Flips |
|---------------|-----|----------|-----------|--------|-------|------------|-------------|------------|-------|
| Anatomy       |     |          |           |        |       |            |             |            |       |
| Mistral Small | 1   | Base     | 0.800     | 0.928  | 0.859 | 128.7      | 7.79        | —          | —     |
|               | 1   | + Critic | 0.815     | 0.915  | 0.862 | 133.1      | 8.24        | 348        | 48    |
|               | 5   | Base     | 0.578     | 0.793  | 0.669 | 88.5       | 3.94        | —          | —     |
|               | 5   | + Critic | 0.610     | 0.754  | 0.674 | 89.7       | 4.05        | 175        | 38    |
| GPT-5.5       | 1   | Base     | 0.951     | 0.978  | 0.965 | 213.3      | 19.98       | —          | —     |
|               | 1   | + Critic | 0.951     | 0.978  | 0.965 | 217.2      | 20.35       | 12         | 0     |
|               | 5   | Base     | 0.796     | 0.950  | 0.866 | 122.5      | 11.52       | —          | —     |
|               | 5   | + Critic | 0.796     | 0.950  | 0.866 | 123.5      | 11.69       | 10         | 0     |
| Conference    |     |          |           |        |       |            |             |            |       |
| Mistral Small | 5   | Base     | 0.290     | 0.469  | 0.358 | 58.0       | 2.86        | —          | —     |
|               | 5   | + Critic | 0.318     | 0.420  | 0.362 | 58.4       | 2.92        | 120        | 24    |
| GPT-5.5       | 5   | Base     | 0.449     | 0.926  | 0.605 | 97.3       | 6.81        | —          | —     |
|               | 5   | + Critic | 0.449     | 0.926  | 0.605 | 97.3       | 6.81        | 17         | 0     |

Mistral Small achieving  $F1 = 0.358$ —a systematic over-matching pattern driven by the short, abstract labels and shallow hierarchies of *Conference* ontologies, which provide insufficient structural evidence for SPARQL-based exploration. Even ontology pairs containing no gold correspondences in the DDM input receive multiple false-positive matches, indicating that the agent struggles to recognise when no valid correspondence exists.

Enabling the *Critic* yields only marginal improvements with Mistral Small: +1.5 pp precision and +0.3 pp F1 on *Anatomy* ( $k=1$ ), and +2.8 pp precision with +0.4 pp F1 on *Conference*—at the cost of −4.9 pp recall. The modest gains suggest that the current revision mechanism, while directionally correct, lacks the discriminative power to substantially alter the *Matcher*’s decisions. For GPT-5.5, the Critic raises between 10 and 17 objections depending on the configuration, yet none result in a decision change (0 flips across all settings), leaving performance identical to the base variant. Given that the critique methodology demonstrably improves Mistral Small’s precision, we attribute this outcome to the current maturity of the agent instructions rather than to a fundamental limitation of the architecture. Refining the propose–critique–revise mechanism to challenge stronger models more effectively remains a key direction for future work.

These results point to a clear finding: model capability currently dominates over the debate mechanism. GPT-5.5 without Critic ( $F1 = 0.605$ ) outperforms Mistral Small with Critic ( $F1 = 0.362$ ) on *Conference* by a wide margin in both precision and recall. Whether this balance shifts as the critique methodology matures remains an open question. Inference cost is substantial, with individual configurations consuming between 88.5M and 217.2M tokens on *Anatomy*—highlighting that efficiency improvements are necessary before the approach scales to larger ontologies.

Table 7 summarises tool usage by GPT-5.5 on the *Anatomy* track ( $k=5$ ). SPARQL queries account for 73.7% of all tool invocations, with the agent issuing approximately eight queries per candidate pair on average. This indicates that DDM primarily relies on active exploration of ontological context rather than surface-level comparison. Retrieval score lookups account for only 1.3% of all tool calls, suggesting that embedding similarity plays only a minor role once deliberation begins. Finally, GPT-5.5 never invokes the code execution tool, suggesting that programmatic computation provides little additional value for the matching process despite being available.

**Table 7**

Tool call statistics for the *Anatomy* track ( $k=5$ ) using GPT-5.5 as *Matching Agent*. The *Critic Agent* raised zero objections in this configuration; reported values reflect a single matching pass without revision.

| Tool                   | Calls         | %    |
|------------------------|---------------|------|
| SPARQL queries         | 16 539        | 73.7 |
| Evidence writing       | 3 087         | 13.8 |
| Evidence reading       | 1 527         | 6.8  |
| Decision persistence   | 995           | 4.4  |
| Retrieval score lookup | 299           | 1.3  |
| <b>Total</b>           | <b>22 447</b> |      |

## 5. Conclusion

This paper presented DRAMA, a novel multi-agent framework for ontology matching that investigates whether structured deliberation can improve alignment decisions for semantically ambiguous correspondences. By combining tool-augmented reasoning with an asymmetric actor–critic architecture, the system introduces iterative, critique-driven revision into the matching process.

The progressive pipeline design proves effective as an efficiency strategy: the two-stage candidate resolution mechanism reduces the search space by 80–91% with near-perfect reject precision ( $>0.997$ ), confirming that expensive reasoning can be selectively allocated to challenging cases. Furthermore, scaling to newer and larger embedding models substantially improves retrieval recall compared with prior work [4], tightening the upper bound for downstream matching.

End-to-end, DRAMA achieves competitive  $F_1$ -scores and recall comparable to the best-performing systems, while a consistent precision gap relative to established matchers remains. Enabling the *Critic* improves precision at the cost of recall; however, a more capable underlying language model currently yields larger gains than the debate mechanism itself. Most strikingly, when GPT-5.5 is used as the *Matching Agent*, the *Critic* raises objections but produces zero decision changes across all configurations—suggesting that an explicit critique mechanism may be redundant when a sufficiently powerful model is employed, and that the value of structured deliberation may diminish as underlying model capability increases.

Several limitations constrain the generalisability of these findings. The evaluation is limited to two OAEI tracks, and the extent to which the architecture generalises to larger and more heterogeneous ontology collections remains an open question. Moreover, the inference cost introduced by the DDM stage represents a significant practical barrier. Running the DDM stage on the *Anatomy* track with GPT-5.5 consumed approximately 185 million tokens and required around 18 hours of wall-clock time for a pair of ontologies totalling roughly 6,000 classes. Extrapolating from these figures under the simplifying assumption that the proportion of candidates forwarded to DDM remains constant, a 10,000-class ontology would already require an estimated 300 million tokens and over one day of computation; a 100,000-class ontology would demand approximately 3 billion tokens and over 12 days of runtime. In practice, costs may grow super-linearly as larger ontologies introduce denser SPARQL traversal and a higher proportion of ambiguous candidates, rendering the current approach infeasible beyond small-to-medium ontologies without substantial efficiency improvements.

Finally, the current system focuses exclusively on class-level equivalence matching and processes source entities independently without global backtracking, which may prevent the identification of globally optimal alignments.

In summary, these results demonstrate that debate-driven architectures represent a feasible, though not yet fully mature, paradigm for ontology matching. Future work will prioritise differentiating objection types to enable more robust revision behaviour, evaluating DRAMA on additional OAEI tracks, and investigating how task framing and contextual information should be structured to optimise both individual agent performance and collaborative effectiveness.

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## Declaration on Generative AI

During the preparation of this work, the author used Claude Opus 4.8 (Anthropic) in order to support the drafting and editing of text passages. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the publication’s content.

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## A. CG: Embedding Model Ablation

Table 8 presents the retrieval recall achieved by seven embedding models on the OAEI *Anatomy* track across retrieval depths  $k \in \{1, \dots, 5\}$ . The evaluated models were selected from the highest-ranked entries on the Hugging Face MTEB embedding leaderboard as of April 28, 2026 [47]. Additionally, `multi-qa-mpnet-base-dot-v1` (110M parameters), one of the embedding models employed by OLaLa [4], was included as a baseline to validate that our retrieval implementation reproduces comparable candidate generation performance under identical retrieval settings.

The results reveal a clear performance gap between the three strongest large-scale embedding models (KaLM-12B, Octen-8B, and Qwen3-8B) and the remaining candidates. All three consistently achieve retrieval recall above 0.96 even at  $k=1$  and outperform the smaller models across all retrieval depths. A notable exception is `llama-embed-nemotron-8b`, which performs substantially below the other large models despite its comparable parameter count. This observation is consistent with its lower

MTEB Retrieval sub-score (68.69 versus 75.66 for KaLM-12B), indicating that strong overall embedding performance does not necessarily translate into superior retrieval effectiveness. Among the top three models, Octen-8B achieves marginally higher recall at  $k=1$  (0.9617 versus 0.9604), whereas KaLM-12B consistently outperforms it for all retrieval depths  $k \geq 2$ .

Recall improvements exhibit clear diminishing returns as the retrieval depth  $k$  increases. For KaLM-12B, the incremental gain decreases from +2.6 percentage points ( $k=1 \rightarrow 2$ ) to +0.4 percentage points ( $k=2 \rightarrow 3$ ) and only +0.1 percentage points ( $k=4 \rightarrow 5$ ), with a similar trend observed across all evaluated models. In particular,  $k=3$  represents an attractive operating point for KaLM-12B, achieving 99.0% recall while leaving only 15 reference correspondences unretrieved. Increasing the retrieval depth from  $k=3$  to  $k=5$  recovers only four additional gold correspondences at the cost of a 63% increase in candidate set size. Moreover, correspondences that are retrieved only for  $k>3$  are consistently ranked low in both retrieval directions, suggesting that they exhibit only weak semantic similarity and therefore provide limited evidence for downstream matching decisions.

Based on these results, KaLM-Embedding-Gemma3-12B-2511 was selected for all subsequent experiments, as it achieves the highest retrieval recall for all  $k \geq 2$  while also providing the fastest runtime among the evaluated large-scale models (263 s versus 360–385 s for the 8B models), despite its larger parameter count.

## B. CR: Hyperparameter Ablation

Table 9 reports the effect of the pruning threshold  $\tau_{\text{low}}$  on candidate set reduction and recall. For both retrieval depths,  $\tau_{\text{low}}=0.71$  constitutes a sharp operating point: all thresholds up to this value incur zero gold loss, while the first reference correspondence is discarded at  $\tau_{\text{low}}=0.72$ .

At this optimal threshold, the candidate set reduction remains moderate—4.4% at  $k=1$  and 11.0% at  $k=5$ —indicating that pruning alone is not the primary driver of search-space reduction; the subsequent acceptance rules and LLM classification contribute the bulk of candidate elimination. More aggressive thresholds offer substantially larger reductions but at increasing recall cost. For instance,  $\tau_{\text{low}}=0.80$  at  $k=5$  eliminates 66.5% of candidates but irreversibly discards 30 reference correspondences, reducing recall from 0.993 to 0.973. This trade-off motivates the conservative choice of  $\tau_{\text{low}}=0.71$ , which preserves the full recall upper bound while delegating further reduction to downstream stages.

Table 10 reports the effect of the acceptance threshold  $\tau_{\text{high}}$  on Stage 1 acceptance behaviour.  $\tau_{\text{high}}=0.97$  emerges as a clear operating point: it yields the highest precision for  $k=1$  (0.9802) and is the lowest threshold achieving perfect precision for  $k=5$  (99 TP, 0 FP).

The trade-off between precision and coverage is substantial. At  $\tau_{\text{high}}=0.80$ , Stage 1 accepts over 1 400 pairs and reduces the candidate set by more than 84%, but precision drops to 0.93 and 50 gold correspondences are irreversibly locked out by incorrect acceptances. At  $\tau_{\text{high}}=0.97$ , only  $\sim 100$  pairs are accepted, limiting reduction to 13–17%, but lock-FN drops to 4 ( $k=1$ ) and 1 ( $k=5$ ), minimising irreversible recall damage.

Notably, at  $k=5$  the additional separation margin  $\delta$  provides an extra safeguard: while  $k=1$  still admits 2 false positives at  $\tau_{\text{high}}=0.97$ , the margin constraint eliminates these entirely at  $k=5$ . This motivates the conservative choice of  $\tau_{\text{high}}=0.97$ , which prioritises near-perfect precision and delegates the remaining candidates to subsequent stages equipped with deeper reasoning.

Table 11 reports the effect of the separation margin  $\delta$  on Stage 1 acceptance, grouped by acceptance threshold  $\tau_{\text{high}}$ .

The interaction between both parameters is pronounced. At lower thresholds ( $\tau_{\text{high}} \in \{0.95, 0.96\}$ ), the margin is essential for eliminating false positives:  $\tau_{\text{high}}=0.95$  requires  $\delta \geq 0.15$  and  $\tau_{\text{high}}=0.96$  requires  $\delta \geq 0.14$  to achieve perfect precision. At  $\tau_{\text{high}} \geq 0.97$ , precision is already perfect without any margin; here,  $\delta$  serves only to eliminate lock-rejected gold correspondences (Lock-FN).

The configuration  $\tau_{\text{high}}=0.97$ ,  $\delta=0.01$  emerges as optimal: it maximises accepted pairs (98 TP, 0 FP) while achieving both perfect precision and zero Lock-FN. In contrast, lower thresholds with higher margins achieve perfect precision at substantially fewer accepts (36–42). Once perfect precision is

reached, further increasing  $\delta$  yields diminishing returns—reducing coverage without additional quality benefit (e.g., at  $\tau_{\text{high}}=0.97$ :  $\delta=0.01 \rightarrow 98$  accepts,  $\delta=0.10 \rightarrow 76$ ,  $\delta=0.20 \rightarrow 3$ ). Results remain stable across  $\delta \in [0.01, 0.05]$ , indicating that the margin has no further discriminative effect in this range.

Notably, Lock-FN = 0 at the selected operating point provides initial evidence that the one-to-one simplification adopted in Stage 1 does not introduce irreversible recall damage on the *Anatomy* track: the separation margin successfully defers cases where competing candidates produce similarly high scores to subsequent stages rather than committing prematurely. Whether this holds across benchmarks with a higher prevalence of m:n correspondences remains to be verified.

**Table 8**

CG retrieval recall by embedding model and  $k$  on the OAEI Anatomy track (1,516 gold correspondences). Alignment Size denotes the number of candidate pairs after bidirectional merge; it varies across models because the set union  $|O_1| \cdot k + |O_2| \cdot k - |\text{overlap}|$  depends on how consistently a model retrieves the same pair from both directions.

| Embedding Model                | Recall        | False Negatives | Size  | Duration (s) |
|--------------------------------|---------------|-----------------|-------|--------------|
| $k = 1$                        |               |                 |       |              |
| multi-qa-mpnet-base-dot-v1     | 0.9307        | 105             | 4606  | 25.93        |
| multilingual-e5-large-instruct | 0.9149        | 129             | 4832  | 37.81        |
| harrier-oss-v1-0.6b            | 0.9294        | 107             | 4752  | 44.18        |
| llama-embed-nemotron-8b        | 0.8562        | 218             | 4882  | 343.94       |
| Octen-Embedding-8B             | <b>0.9617</b> | 58              | 4551  | 363.86       |
| Qwen3-Embedding-8B             | 0.9571        | 65              | 4577  | 360.37       |
| KaLM-Embedding-Gemma3-12B-2511 | 0.9604        | 60              | 4560  | 263.36       |
| $k = 2$                        |               |                 |       |              |
| multi-qa-mpnet-base-dot-v1     | 0.9551        | 68              | 9396  | 32.25        |
| multilingual-e5-large-instruct | 0.9538        | 70              | 9775  | 45.26        |
| harrier-oss-v1-0.6b            | 0.9512        | 74              | 9715  | 51.93        |
| llama-embed-nemotron-8b        | 0.8879        | 170             | 9684  | 363.27       |
| Octen-Embedding-8B             | 0.9796        | 31              | 9293  | 383.82       |
| Qwen3-Embedding-8B             | 0.9723        | 42              | 9355  | 380.03       |
| KaLM-Embedding-Gemma3-12B-2511 | <b>0.9861</b> | 21              | 9317  | 281.93       |
| $k = 3$                        |               |                 |       |              |
| multi-qa-mpnet-base-dot-v1     | 0.9650        | 53              | 13968 | 32.78        |
| multilingual-e5-large-instruct | 0.9677        | 49              | 14576 | 45.78        |
| harrier-oss-v1-0.6b            | 0.9617        | 58              | 14445 | 52.31        |
| llama-embed-nemotron-8b        | 0.9070        | 141             | 14286 | 363.98       |
| Octen-Embedding-8B             | 0.9868        | 20              | 13752 | 384.12       |
| Qwen3-Embedding-8B             | 0.9782        | 33              | 13876 | 380.56       |
| KaLM-Embedding-Gemma3-12B-2511 | <b>0.9901</b> | 15              | 13807 | 282.64       |
| $k = 4$                        |               |                 |       |              |
| multi-qa-mpnet-base-dot-v1     | 0.9710        | 44              | 18495 | 33.44        |
| multilingual-e5-large-instruct | 0.9736        | 40              | 19330 | 46.64        |
| harrier-oss-v1-0.6b            | 0.9782        | 33              | 19100 | 52.83        |
| llama-embed-nemotron-8b        | 0.9162        | 127             | 18823 | 364.50       |
| Octen-Embedding-8B             | 0.9875        | 19              | 18109 | 384.62       |
| Qwen3-Embedding-8B             | 0.9829        | 26              | 18300 | 380.97       |
| KaLM-Embedding-Gemma3-12B-2511 | <b>0.9914</b> | 13              | 18216 | 282.60       |
| $k = 5$                        |               |                 |       |              |
| multi-qa-mpnet-base-dot-v1     | 0.9776        | 34              | 22999 | 33.57        |
| multilingual-e5-large-instruct | 0.9763        | 36              | 24028 | 46.15        |
| harrier-oss-v1-0.6b            | 0.9809        | 29              | 23748 | 53.15        |
| llama-embed-nemotron-8b        | 0.9241        | 115             | 23258 | 364.98       |
| Octen-Embedding-8B             | 0.9908        | 14              | 22377 | 384.80       |
| Qwen3-Embedding-8B             | 0.9835        | 25              | 22729 | 381.18       |
| KaLM-Embedding-Gemma3-12B-2511 | <b>0.9927</b> | 11              | 22540 | 282.78       |

**Table 9**

Effect of the **pruning threshold**  $\tau_{\text{low}}$  on CR Stage 1 filtering for KaLM-Embedding-Gemma3-12B-2511 with different retrieval depths on the OAEI Anatomy track (1,516 gold correspondences). Recall denotes the post-prune upper bound, i.e. the fraction of gold correspondences still reachable by downstream stages. The highlighted rows mark the highest threshold that incurs zero gold loss for each retrieval depth.

| $\tau_{\text{low}}$ | Pruned       | Pruned Gold | Recall        | Recall Loss   | Alignment Reduction | Size          |
|---------------------|--------------|-------------|---------------|---------------|---------------------|---------------|
| $k=1$               |              |             |               |               |                     |               |
| 0.60                | 0            | 0           | 0.9604        | 0.0000        | 0.00%               | 4 560         |
| 0.61                | 0            | 0           | 0.9604        | 0.0000        | 0.00%               | 4 560         |
| 0.62                | 0            | 0           | 0.9604        | 0.0000        | 0.00%               | 4 560         |
| 0.63                | 1            | 0           | 0.9604        | 0.0000        | 0.02%               | 4 559         |
| 0.64                | 6            | 0           | 0.9604        | 0.0000        | 0.13%               | 4 554         |
| 0.65                | 23           | 0           | 0.9604        | 0.0000        | 0.50%               | 4 537         |
| 0.66                | 40           | 0           | 0.9604        | 0.0000        | 0.88%               | 4 520         |
| 0.67                | 59           | 0           | 0.9604        | 0.0000        | 1.29%               | 4 501         |
| 0.68                | 81           | 0           | 0.9604        | 0.0000        | 1.78%               | 4 479         |
| 0.69                | 114          | 0           | 0.9604        | 0.0000        | 2.50%               | 4 446         |
| 0.70                | 150          | 0           | 0.9604        | 0.0000        | 3.29%               | 4 410         |
| <b>0.71</b>         | <b>201</b>   | <b>0</b>    | <b>0.9604</b> | <b>0.0000</b> | <b>4.41%</b>        | <b>4 359</b>  |
| 0.72                | 255          | 1           | 0.9598        | 0.0007        | 5.59%               | 4 305         |
| 0.73                | 337          | 1           | 0.9598        | 0.0007        | 7.39%               | 4 223         |
| 0.74                | 439          | 1           | 0.9598        | 0.0007        | 9.63%               | 4 121         |
| 0.75                | 551          | 1           | 0.9598        | 0.0007        | 12.08%              | 4 009         |
| 0.76                | 717          | 1           | 0.9598        | 0.0007        | 15.72%              | 3 843         |
| 0.77                | 910          | 5           | 0.9571        | 0.0033        | 19.96%              | 3 650         |
| 0.78                | 1 116        | 7           | 0.9558        | 0.0046        | 24.47%              | 3 444         |
| 0.79                | 1 314        | 7           | 0.9558        | 0.0046        | 28.82%              | 3 246         |
| 0.80                | 1 504        | 15          | 0.9505        | 0.0099        | 32.98%              | 3 056         |
| $k=5$               |              |             |               |               |                     |               |
| 0.60                | 0            | 0           | 0.9927        | 0.0000        | 0.00%               | 22 540        |
| 0.61                | 3            | 0           | 0.9927        | 0.0000        | 0.01%               | 22 537        |
| 0.62                | 22           | 0           | 0.9927        | 0.0000        | 0.10%               | 22 518        |
| 0.63                | 86           | 0           | 0.9927        | 0.0000        | 0.38%               | 22 454        |
| 0.64                | 189          | 0           | 0.9927        | 0.0000        | 0.84%               | 22 351        |
| 0.65                | 331          | 0           | 0.9927        | 0.0000        | 1.47%               | 22 209        |
| 0.66                | 484          | 0           | 0.9927        | 0.0000        | 2.15%               | 22 056        |
| 0.67                | 683          | 0           | 0.9927        | 0.0000        | 3.03%               | 21 857        |
| 0.68                | 973          | 0           | 0.9927        | 0.0000        | 4.32%               | 21 567        |
| 0.69                | 1 336        | 0           | 0.9927        | 0.0000        | 5.93%               | 21 204        |
| 0.70                | 1 824        | 0           | 0.9927        | 0.0000        | 8.09%               | 20 716        |
| <b>0.71</b>         | <b>2 476</b> | <b>0</b>    | <b>0.9927</b> | <b>0.0000</b> | <b>10.98%</b>       | <b>20 064</b> |
| 0.72                | 3 317        | 1           | 0.9921        | 0.0007        | 14.72%              | 19 223        |
| 0.73                | 4 399        | 2           | 0.9914        | 0.0013        | 19.52%              | 18 141        |
| 0.74                | 5 718        | 3           | 0.9907        | 0.0020        | 25.37%              | 16 822        |
| 0.75                | 7 134        | 6           | 0.9888        | 0.0040        | 31.65%              | 15 406        |
| 0.76                | 8 783        | 7           | 0.9881        | 0.0046        | 38.97%              | 13 757        |
| 0.77                | 10 470       | 15          | 0.9828        | 0.0099        | 46.45%              | 12 070        |
| 0.78                | 12 092       | 19          | 0.9802        | 0.0125        | 53.65%              | 10 448        |
| 0.79                | 13 638       | 22          | 0.9782        | 0.0145        | 60.51%              | 8 902         |
| 0.80                | 14 994       | 30          | 0.9729        | 0.0198        | 66.52%              | 7 546         |

**Table 10**

Effect of the **acceptance threshold**  $\tau_{\text{high}}$  on Stage 1 acceptance and workload reduction for KaLM-Embedding-Gemma3-12B-2511 with different retrieval depths on the OAEI Anatomy track (1,516 gold correspondences).  $\tau_{\text{low}}=0.71$  and  $\delta=0.00$  are fixed. Alignment Size denotes the number of pairs forwarded to Stage 2. Reduction metrics include lock-rejected pairs. The highlighted rows mark the configurations with the highest precision for each retrieval depth.

| $\tau_{\text{high}}$ | Accepts    | TP        | FP       | Precision     | Lock-FN  | Size          | $\tau_{\text{high}}$ Reduction | Total Reduction |
|----------------------|------------|-----------|----------|---------------|----------|---------------|--------------------------------|-----------------|
| $k=1$                |            |           |          |               |          |               |                                |                 |
| 0.80                 | 1 448      | 1 343     | 105      | 0.9275        | 50       | 524           | 84.10%                         | 88.51%          |
| 0.81                 | 1 437      | 1 340     | 97       | 0.9325        | 49       | 555           | 83.42%                         | 87.83%          |
| 0.82                 | 1 427      | 1 339     | 88       | 0.9383        | 49       | 581           | 82.85%                         | 87.26%          |
| 0.83                 | 1 416      | 1 333     | 83       | 0.9414        | 49       | 607           | 82.28%                         | 86.69%          |
| 0.84                 | 1 392      | 1 317     | 75       | 0.9461        | 49       | 656           | 81.21%                         | 85.61%          |
| 0.85                 | 1 374      | 1 306     | 68       | 0.9505        | 48       | 696           | 80.33%                         | 84.74%          |
| 0.86                 | 1 348      | 1 289     | 59       | 0.9562        | 47       | 753           | 79.08%                         | 83.49%          |
| 0.87                 | 1 324      | 1 269     | 55       | 0.9585        | 47       | 796           | 78.14%                         | 82.54%          |
| 0.88                 | 1 282      | 1 233     | 49       | 0.9618        | 44       | 895           | 75.96%                         | 80.37%          |
| 0.89                 | 1 237      | 1 196     | 41       | 0.9669        | 42       | 993           | 73.82%                         | 78.22%          |
| 0.90                 | 1 181      | 1 150     | 31       | 0.9738        | 38       | 1 119         | 71.05%                         | 75.46%          |
| 0.91                 | 1 120      | 1 091     | 29       | 0.9741        | 37       | 1 278         | 67.57%                         | 71.97%          |
| 0.92                 | 1 016      | 991       | 25       | 0.9754        | 35       | 1 576         | 61.03%                         | 65.44%          |
| 0.93                 | 843        | 821       | 22       | 0.9739        | 31       | 2 015         | 51.40%                         | 55.81%          |
| 0.94                 | 564        | 546       | 18       | 0.9681        | 25       | 2 646         | 37.57%                         | 41.97%          |
| 0.95                 | 323        | 314       | 9        | 0.9721        | 14       | 3 231         | 24.74%                         | 29.14%          |
| 0.96                 | 167        | 163       | 4        | 0.9760        | 7        | 3 585         | 16.97%                         | 21.38%          |
| <b>0.97</b>          | <b>101</b> | <b>99</b> | <b>2</b> | <b>0.9802</b> | <b>4</b> | <b>3 753</b>  | <b>13.29%</b>                  | <b>17.70%</b>   |
| 0.98                 | 94         | 92        | 2        | 0.9787        | 4        | 3 800         | 12.26%                         | 16.67%          |
| 0.99                 | 88         | 86        | 2        | 0.9773        | 4        | 3 824         | 11.73%                         | 16.14%          |
| 1.00                 | 53         | 51        | 2        | 0.9623        | 4        | 4 080         | 6.12%                          | 10.53%          |
| $k=5$                |            |           |          |               |          |               |                                |                 |
| 0.80                 | 1 438      | 1 340     | 98       | 0.9318        | 55       | 3 260         | 74.55%                         | 85.54%          |
| 0.81                 | 1 427      | 1 337     | 90       | 0.9369        | 53       | 3 341         | 74.19%                         | 85.18%          |
| 0.82                 | 1 417      | 1 336     | 81       | 0.9428        | 51       | 3 441         | 73.75%                         | 84.73%          |
| 0.83                 | 1 406      | 1 330     | 76       | 0.9459        | 51       | 3 513         | 73.43%                         | 84.41%          |
| 0.84                 | 1 382      | 1 314     | 68       | 0.9508        | 49       | 3 699         | 72.60%                         | 83.59%          |
| 0.85                 | 1 365      | 1 303     | 62       | 0.9546        | 48       | 3 829         | 72.03%                         | 83.01%          |
| 0.86                 | 1 340      | 1 286     | 54       | 0.9597        | 48       | 4 015         | 71.20%                         | 82.19%          |
| 0.87                 | 1 316      | 1 266     | 50       | 0.9620        | 48       | 4 197         | 70.39%                         | 81.38%          |
| 0.88                 | 1 275      | 1 230     | 45       | 0.9647        | 44       | 4 557         | 68.80%                         | 79.78%          |
| 0.89                 | 1 230      | 1 193     | 37       | 0.9699        | 41       | 4 957         | 67.02%                         | 78.01%          |
| 0.90                 | 1 174      | 1 147     | 27       | 0.9770        | 35       | 5 456         | 64.81%                         | 75.79%          |
| 0.91                 | 1 113      | 1 088     | 25       | 0.9775        | 34       | 6 023         | 62.29%                         | 73.28%          |
| 0.92                 | 1 009      | 988       | 21       | 0.9792        | 32       | 7 136         | 57.36%                         | 68.34%          |
| 0.93                 | 836        | 818       | 18       | 0.9785        | 27       | 9 095         | 48.66%                         | 59.65%          |
| 0.94                 | 559        | 543       | 16       | 0.9714        | 21       | 12 276        | 34.55%                         | 45.54%          |
| 0.95                 | 319        | 312       | 7        | 0.9781        | 10       | 15 278        | 21.23%                         | 32.22%          |
| 0.96                 | 165        | 163       | 2        | 0.9879        | 4        | 16 999        | 13.60%                         | 24.58%          |
| <b>0.97</b>          | <b>99</b>  | <b>99</b> | <b>0</b> | <b>1.0000</b> | <b>1</b> | <b>17 833</b> | <b>9.90%</b>                   | <b>20.88%</b>   |
| 0.98                 | 92         | 92        | 0        | 1.0000        | 1        | 17 984        | 9.23%                          | 20.21%          |
| 0.99                 | 86         | 86        | 0        | 1.0000        | 1        | 18 090        | 8.76%                          | 19.74%          |
| 1.00                 | 51         | 51        | 0        | 1.0000        | 1        | 18 956        | 4.92%                          | 15.90%          |

**Table 11**

Effect of the separation margin  $\delta$  on Stage 1 acceptance for KaLM-Embedding-Gemma3-12B-2511 with  $k=5$  and  $\tau_{\text{low}}=0.71$  on the OAEI Anatomy track (1,516 gold correspondences; 20,064 candidate pairs after pruning). The highlighted row marks the lowest  $\delta$  that achieves perfect precision. Results are grouped by acceptance threshold  $\tau_{\text{high}}$ .

| $\delta$                    | Accepts   | TP        | FP       | Precision     | Lock-FN  | Size          | $\tau_{\text{high}}$ Reduction | Total Reduction |
|-----------------------------|-----------|-----------|----------|---------------|----------|---------------|--------------------------------|-----------------|
| $\tau_{\text{high}} = 0.95$ |           |           |          |               |          |               |                                |                 |
| 0.10                        | 156       | 153       | 3        | 0.9808        | 0        | 17 476        | 11.48%                         | 22.47%          |
| 0.11                        | 123       | 121       | 2        | 0.9837        | 0        | 18 133        | 8.57%                          | 19.55%          |
| 0.12                        | 92        | 90        | 2        | 0.9783        | 0        | 18 651        | 6.27%                          | 17.25%          |
| 0.13                        | 72        | 70        | 2        | 0.9722        | 0        | 19 011        | 4.67%                          | 15.66%          |
| 0.14                        | 52        | 51        | 1        | 0.9808        | 0        | 19 327        | 3.27%                          | 14.25%          |
| <b>0.15</b>                 | <b>42</b> | <b>42</b> | <b>0</b> | <b>1.0000</b> | <b>0</b> | <b>19 453</b> | <b>2.71%</b>                   | <b>13.70%</b>   |
| 0.16                        | 33        | 33        | 0        | 1.0000        | 0        | 19 632        | 1.92%                          | 12.90%          |
| 0.17                        | 24        | 24        | 0        | 1.0000        | 0        | 19 752        | 1.38%                          | 12.37%          |
| 0.18                        | 16        | 16        | 0        | 1.0000        | 0        | 19 863        | 0.89%                          | 11.88%          |
| 0.19                        | 8         | 8         | 0        | 1.0000        | 0        | 19 979        | 0.38%                          | 11.36%          |
| 0.20                        | 5         | 5         | 0        | 1.0000        | 0        | 20 025        | 0.17%                          | 11.16%          |
| $\tau_{\text{high}} = 0.96$ |           |           |          |               |          |               |                                |                 |
| 0.10                        | 102       | 101       | 1        | 0.9902        | 0        | 18 111        | 8.66%                          | 19.65%          |
| 0.11                        | 85        | 84        | 1        | 0.9882        | 0        | 18 574        | 6.61%                          | 17.60%          |
| 0.12                        | 66        | 65        | 1        | 0.9848        | 0        | 18 957        | 4.91%                          | 15.90%          |
| 0.13                        | 47        | 46        | 1        | 0.9787        | 0        | 19 306        | 3.36%                          | 14.35%          |
| <b>0.14</b>                 | <b>36</b> | <b>36</b> | <b>0</b> | <b>1.0000</b> | <b>0</b> | <b>19 513</b> | <b>2.44%</b>                   | <b>13.43%</b>   |
| 0.15                        | 30        | 30        | 0        | 1.0000        | 0        | 19 588        | 2.11%                          | 13.10%          |
| 0.16                        | 23        | 23        | 0        | 1.0000        | 0        | 19 723        | 1.51%                          | 12.50%          |
| 0.17                        | 15        | 15        | 0        | 1.0000        | 0        | 19 835        | 1.02%                          | 12.00%          |
| 0.18                        | 9         | 9         | 0        | 1.0000        | 0        | 19 927        | 0.61%                          | 11.59%          |
| 0.19                        | 5         | 5         | 0        | 1.0000        | 0        | 20 001        | 0.28%                          | 11.26%          |
| 0.20                        | 3         | 3         | 0        | 1.0000        | 0        | 20 036        | 0.12%                          | 11.11%          |
| $\tau_{\text{high}} = 0.97$ |           |           |          |               |          |               |                                |                 |
| 0.00                        | 99        | 99        | 0        | 1.0000        | 1        | 17 833        | 9.90%                          | 20.88%          |
| <b>0.01</b>                 | <b>98</b> | <b>98</b> | <b>0</b> | <b>1.0000</b> | <b>0</b> | <b>17 853</b> | <b>9.81%</b>                   | <b>20.79%</b>   |
| 0.02                        | 98        | 98        | 0        | 1.0000        | 0        | 17 853        | 9.81%                          | 20.79%          |
| 0.03                        | 98        | 98        | 0        | 1.0000        | 0        | 17 853        | 9.81%                          | 20.79%          |
| 0.04                        | 98        | 98        | 0        | 1.0000        | 0        | 17 853        | 9.81%                          | 20.79%          |
| 0.05                        | 98        | 98        | 0        | 1.0000        | 0        | 17 853        | 9.81%                          | 20.79%          |
| 0.06                        | 96        | 96        | 0        | 1.0000        | 0        | 17 888        | 9.65%                          | 20.64%          |
| 0.07                        | 95        | 95        | 0        | 1.0000        | 0        | 17 900        | 9.60%                          | 20.59%          |
| 0.08                        | 91        | 91        | 0        | 1.0000        | 0        | 18 008        | 9.12%                          | 20.11%          |
| 0.09                        | 82        | 82        | 0        | 1.0000        | 0        | 18 266        | 7.98%                          | 18.96%          |
| 0.10                        | 76        | 76        | 0        | 1.0000        | 0        | 18 438        | 7.21%                          | 18.20%          |
| 0.15                        | 22        | 22        | 0        | 1.0000        | 0        | 19 694        | 1.64%                          | 12.63%          |
| 0.20                        | 3         | 3         | 0        | 1.0000        | 0        | 20 036        | 0.12%                          | 11.11%          |

| $\delta$                    | Accepts   | TP        | FP       | Precision     | Lock-FN  | Alignment Size | $\tau_{\text{high}}$ Reduction | Total Reduction |
|-----------------------------|-----------|-----------|----------|---------------|----------|----------------|--------------------------------|-----------------|
| $\tau_{\text{high}} = 0.98$ |           |           |          |               |          |                |                                |                 |
| 0.00                        | 92        | 92        | 0        | 1.0000        | 1        | 17 984         | 9.23%                          | 20.21%          |
| <b>0.01</b>                 | <b>91</b> | <b>91</b> | <b>0</b> | <b>1.0000</b> | <b>0</b> | <b>18 004</b>  | <b>9.14%</b>                   | <b>20.12%</b>   |
| 0.02                        | 91        | 91        | 0        | 1.0000        | 0        | 18 004         | 9.14%                          | 20.12%          |
| 0.03                        | 91        | 91        | 0        | 1.0000        | 0        | 18 004         | 9.14%                          | 20.12%          |
| 0.04                        | 91        | 91        | 0        | 1.0000        | 0        | 18 004         | 9.14%                          | 20.12%          |
| 0.05                        | 91        | 91        | 0        | 1.0000        | 0        | 18 004         | 9.14%                          | 20.12%          |
| 0.06                        | 89        | 89        | 0        | 1.0000        | 0        | 18 039         | 8.98%                          | 19.97%          |
| 0.07                        | 89        | 89        | 0        | 1.0000        | 0        | 18 039         | 8.98%                          | 19.97%          |
| 0.08                        | 86        | 86        | 0        | 1.0000        | 0        | 18 137         | 8.55%                          | 19.53%          |
| 0.09                        | 80        | 80        | 0        | 1.0000        | 0        | 18 301         | 7.82%                          | 18.81%          |
| 0.10                        | 74        | 74        | 0        | 1.0000        | 0        | 18 473         | 7.06%                          | 18.04%          |
| 0.15                        | 22        | 22        | 0        | 1.0000        | 0        | 19 694         | 1.64%                          | 12.63%          |
| 0.20                        | 3         | 3         | 0        | 1.0000        | 0        | 20 036         | 0.12%                          | 11.11%          |
| $\tau_{\text{high}} = 0.99$ |           |           |          |               |          |                |                                |                 |
| 0.00                        | 86        | 86        | 0        | 1.0000        | 1        | 18 090         | 8.76%                          | 19.74%          |
| <b>0.01</b>                 | <b>85</b> | <b>85</b> | <b>0</b> | <b>1.0000</b> | <b>0</b> | <b>18 110</b>  | <b>8.67%</b>                   | <b>19.65%</b>   |
| 0.02                        | 85        | 85        | 0        | 1.0000        | 0        | 18 110         | 8.67%                          | 19.65%          |
| 0.03                        | 85        | 85        | 0        | 1.0000        | 0        | 18 110         | 8.67%                          | 19.65%          |
| 0.04                        | 85        | 85        | 0        | 1.0000        | 0        | 18 110         | 8.67%                          | 19.65%          |
| 0.05                        | 85        | 85        | 0        | 1.0000        | 0        | 18 110         | 8.67%                          | 19.65%          |
| 0.06                        | 85        | 85        | 0        | 1.0000        | 0        | 18 110         | 8.67%                          | 19.65%          |
| 0.07                        | 85        | 85        | 0        | 1.0000        | 0        | 18 110         | 8.67%                          | 19.65%          |
| 0.08                        | 82        | 82        | 0        | 1.0000        | 0        | 18 208         | 8.23%                          | 19.22%          |
| 0.09                        | 77        | 77        | 0        | 1.0000        | 0        | 18 358         | 7.57%                          | 18.55%          |
| 0.10                        | 71        | 71        | 0        | 1.0000        | 0        | 18 530         | 6.81%                          | 17.79%          |
| 0.15                        | 22        | 22        | 0        | 1.0000        | 0        | 19 694         | 1.64%                          | 12.63%          |
| 0.20                        | 3         | 3         | 0        | 1.0000        | 0        | 20 036         | 0.12%                          | 11.11%          |
| $\tau_{\text{high}} = 1.00$ |           |           |          |               |          |                |                                |                 |
| 0.00                        | 51        | 51        | 0        | 1.0000        | 1        | 18 956         | 4.92%                          | 15.90%          |
| <b>0.01</b>                 | <b>50</b> | <b>50</b> | <b>0</b> | <b>1.0000</b> | <b>0</b> | <b>18 980</b>  | <b>4.81%</b>                   | <b>15.79%</b>   |
| 0.02                        | 50        | 50        | 0        | 1.0000        | 0        | 18 980         | 4.81%                          | 15.79%          |
| 0.03                        | 50        | 50        | 0        | 1.0000        | 0        | 18 980         | 4.81%                          | 15.79%          |
| 0.04                        | 50        | 50        | 0        | 1.0000        | 0        | 18 980         | 4.81%                          | 15.79%          |
| 0.05                        | 50        | 50        | 0        | 1.0000        | 0        | 18 980         | 4.81%                          | 15.79%          |
| 0.06                        | 50        | 50        | 0        | 1.0000        | 0        | 18 980         | 4.81%                          | 15.79%          |
| 0.07                        | 50        | 50        | 0        | 1.0000        | 0        | 18 980         | 4.81%                          | 15.79%          |
| 0.08                        | 48        | 48        | 0        | 1.0000        | 0        | 19 029         | 4.59%                          | 15.58%          |
| 0.09                        | 44        | 44        | 0        | 1.0000        | 0        | 19 165         | 3.99%                          | 14.97%          |
| 0.10                        | 39        | 39        | 0        | 1.0000        | 0        | 19 313         | 3.33%                          | 14.32%          |
| 0.15                        | 15        | 15        | 0        | 1.0000        | 0        | 19 819         | 1.09%                          | 12.07%          |
| 0.20                        | 3         | 3         | 0        | 1.0000        | 0        | 20 036         | 0.12%                          | 11.11%          |